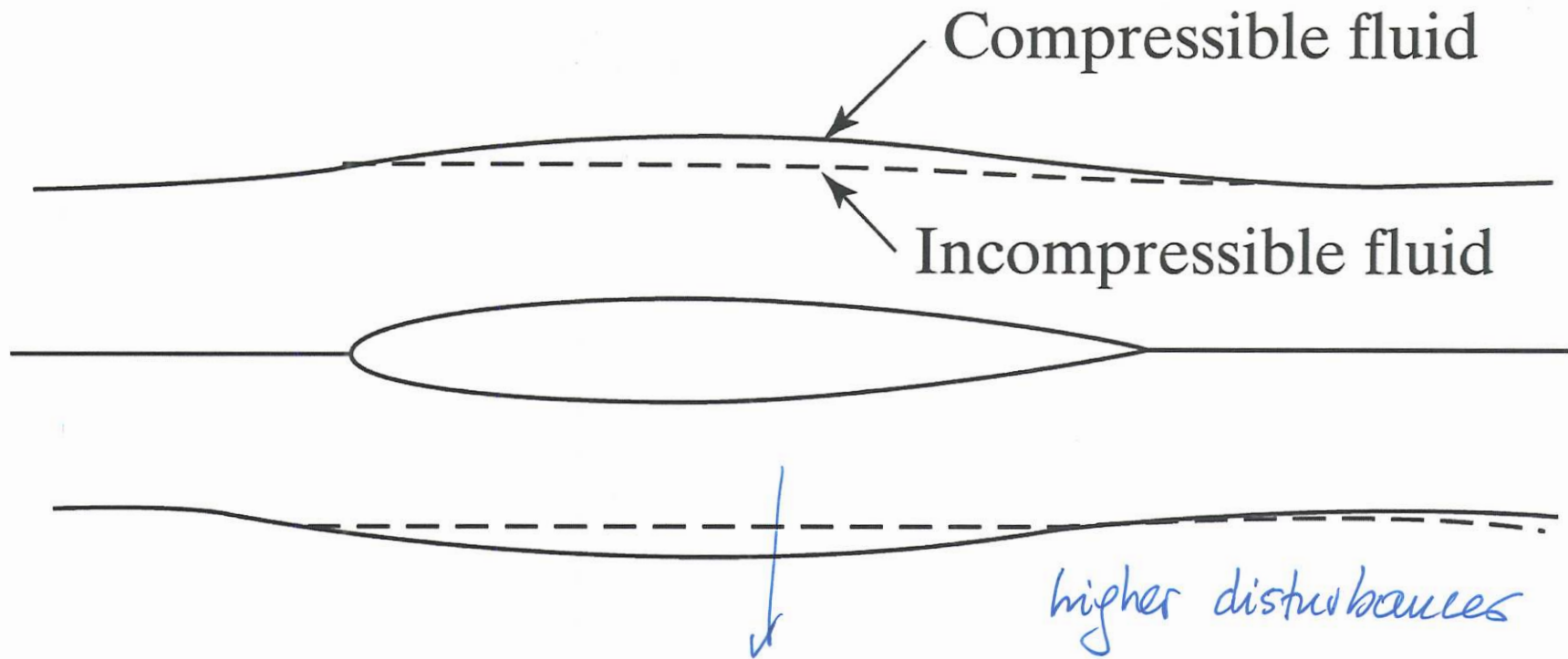


Subsonic Compressible & Transonic Flows [chapter 9]

- Study impact of compressibility to governing equations
- Transform incompressible experimental data or geometry to subsonic compressible Mach numbers. (compressibility "corrections")
- Introduction to transonic flow and its effects.
- Study critical and drag divergence Mach number.
- Introduction to supercritical airfoils, transonic area rule, wing sweep.



A comparison between compressible and incompressible flow shows compressible streamlines opening more and disturbances propagate to ^a larger distance.

Larger speeds lead to critical Mach number ³ and transonic flows

Compressible Subsonic Flow

Inviscid, compressible, subsonic or supersonic (no strong shocks), flow over a body immersed in a uniform stream is *Irrotational* (no mechanism for rotation) $\Rightarrow$ velocity potential can be defined

Assume: 2-D ^{-for simplicity} [text has 3D], steady, irrotational, isentropic flow

define $\phi(x, y)$: [irrotational flows are isentropic] and isentropic flows are irrotational

$$\mathbf{V} = \nabla \phi$$

$$\Rightarrow u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}$$

potential flow approximation OK for weak shocks

e. g. if $M_1 = 1 + \epsilon$, ϵ is small can prove that $s_2 - s_1 = O(\epsilon)^3$ (small)
across a shock

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \quad \Rightarrow \quad \begin{array}{l} \text{steady 2-D} \\ \text{continuity equation} \end{array} \quad (9.1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (9.2a)$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad (9.2b)$$

*x - direction
conservation of
momentum
y - direction*

$$a^2 = \left(\frac{\partial p}{\partial \rho} \right)_s \quad (8.28) \quad (9.3)$$

Combining (9.1)-(9.3) for isentropic flow

$$\left(1 - \frac{u^2}{a^2} \right) \frac{\partial u}{\partial x} + \left(1 - \frac{v^2}{a^2} \right) \frac{\partial v}{\partial y} - \frac{2uv}{a^2} \frac{\partial u}{\partial y} = 0 \quad (9.4a)$$

$$\left(1 - \frac{\phi_x^2}{a^2}\right)\phi_{xx} + \left(1 - \frac{\phi_y^2}{a^2}\right)\phi_{yy} - \frac{\phi_x\phi_y}{a^2}\phi_{xy} = 0 \quad \text{1 non-linear eq.} \quad (9.4b)$$

full velocity potential equation (only ϕ and a appears)

a can be expressed in terms of ϕ :

$$\left(\phi_x = \frac{\partial \phi}{\partial x}, \phi_y = \frac{\partial \phi}{\partial y}\right)$$

$$\frac{a^2}{a_\infty^2} = 1 - \frac{\gamma - 1}{2} \left(\frac{u^2 + v^2}{U_\infty^2} - 1 \right) M_\infty^2 \Rightarrow \frac{a^2}{a_\infty^2} = -\frac{\gamma - 1}{2} \left(\frac{\phi_x^2 + \phi_y^2}{U_\infty^2} - 1 \right) M_\infty^2 \quad (9.6)$$

Eq. (9.4) represents continuity momentum & energy can be easily

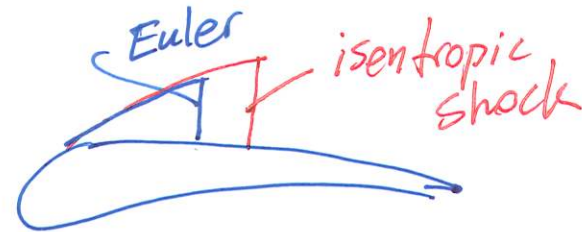
(CFD!) solved (only 1 equation); find ϕ then find u, v, a ; $M = \frac{U}{a}$; $T,$

p, ρ from isentropic flow relations (isentropic tables) can be used to

represent weak shocks (transonic flow)

- not very strong shocks ($\Rightarrow$ isentropic)

[isentropic shocks are stronger]



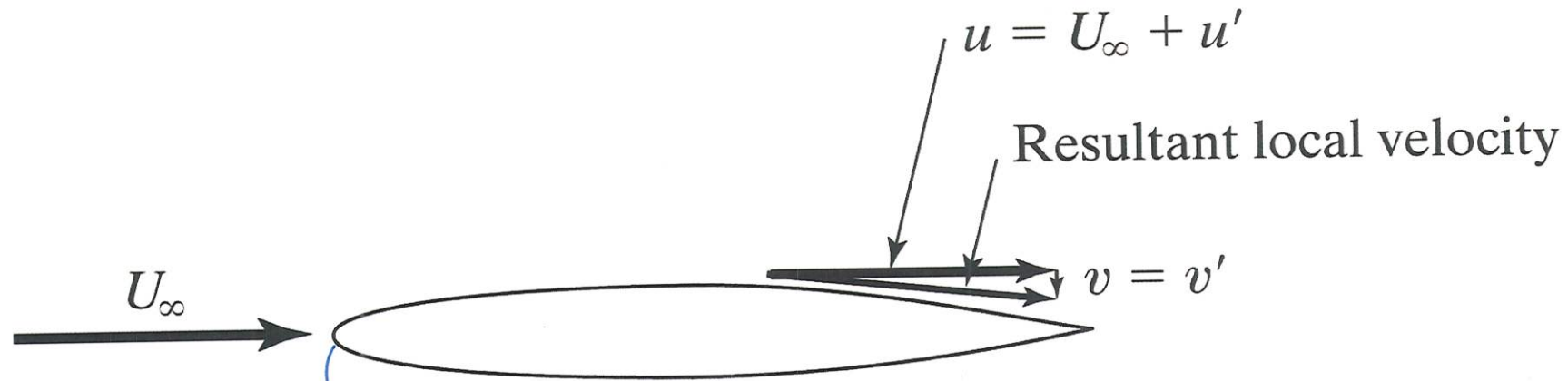
- entropy corrections can be added to the model.

Next level-up Euler eqs. (more expensive)

Still eq. (^{9.4}~~5.12~~) is nonlinear; solve computationally (finite difference methods) $\rightarrow$ used a lot in industry (e.g. flow around transonic wings)

linear approximations (subsonic – supersonic)

The linearized velocity potential equation



Let u' and v' be perturbations of the velocity in the x and y directions

Perturbation velocities u' and v' can be large near the leading edge

stagnation region.

$$|u| \sim |v_\infty|$$

$$u = U_\infty + u'; \quad v = v'$$

ζ_p might not be
accurate near (10%)
leading edge
(similar to thin airfoil
theory)

-but lift is ok!

From (9.4a) for small perturbations

$$\left(1 - \frac{u^2}{a^2}\right) \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (9.5)$$

Extra assumptions:

(go fast over
- algebra 9.6 - 9.10)

From (9.6) using binomial expansion for small perturbations

$$\frac{a_\infty^2}{a^2} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \left(2 \frac{u'}{U_\infty} + \frac{u'^2 + v'^2}{U_\infty^2} \right) \quad (9.7)$$

To simplify (9.5): note that

$$1 - \frac{u^2}{a^2} = 1 - \frac{U_\infty^2 + 2u'U_\infty + u'^2}{U_\infty^2} M_\infty^2 \frac{a_\infty^2}{a^2} \quad (9.8)$$

($u = U_\infty + u'$)

Substituting (9.7) to (9.8) and neglecting high-order terms

$$1 - \frac{u^2}{a^2} = 1 - M_\infty^2 \left[1 + \frac{2u'}{U_\infty} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \right] \quad (9.9)$$

Substitute (9.9) to (9.5)

$$(1 - M_\infty^2) \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = M_\infty^2 \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \frac{2u'}{U_\infty} \frac{\partial u}{\partial x}$$

Using disturbances:

$$(1 - M_\infty^2) \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = \frac{2}{U_\infty} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) M_\infty^2 u' \frac{\partial u'}{\partial x} \quad (9.10)$$

Right hand side can be neglected if *disturbances are small*

$$\Rightarrow \left(1 - M_{\infty}^2\right) \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = 0 \quad (9.11)$$

Perturbation potential ϕ' :

$$\phi = U_{\infty} x + \phi'$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = U_{\infty} + \frac{\partial \phi'}{\partial x}, \quad \frac{\partial \phi}{\partial y} = \frac{\partial \phi'}{\partial y}$$

$$\left(1 - M_{\infty}^2\right) \phi_{xx} + \phi_{yy} = 0 \Rightarrow \underline{\beta^2 \phi_{xx} + \phi_{yy} = 0} \quad (9.13)$$

where $\beta \equiv \sqrt{1 - M_{\infty}^2}$

Prandtl Glauert equation

Assumptions:

- 1) small perturbations (thin bodies, small α)
- 2) subsonic & supersonic M_∞

Not good for: thick bodies or large α

transonic flow $0.8 < M_\infty < 1.2$

hypersonic flow $M_\infty > 5$

hierarchy:

Navier Stokes; Euler; full potential; TSD; linear potential

The Gothert Transformation

Objective: transform (9.13) to Laplace's equation:

$$x' = \frac{x}{\sqrt{1-M_\infty^2}} = \frac{x}{\beta} \quad y' = y \quad (9.14)$$

Then (9.13) becomes: $\phi_{x'x'} + \phi_{y'y'} = 0$ (9.15)

Disturbance pressure coefficient

$$C_p = -\frac{2u'}{U_\infty} = -\frac{2\phi'_x}{U_\infty}$$

(linear approximation)
- similar to thin airfoil theory
depends only on u' (not v')

(9.16a)

Therefore:

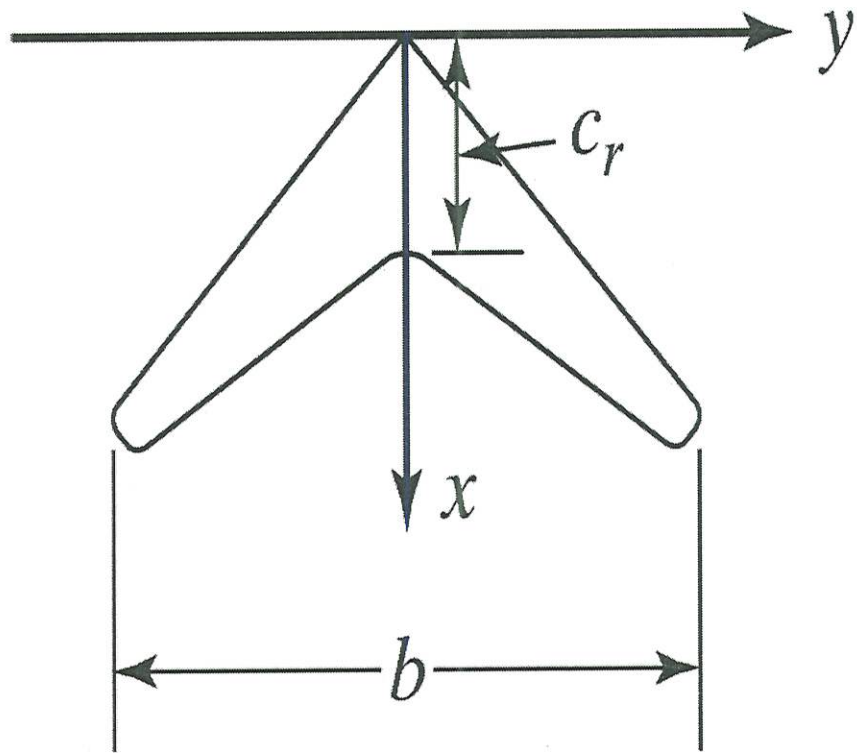
$$C_p = -\frac{C'_p}{\beta}$$

(9.16b)

Prandtl-Glauert compressibility correction

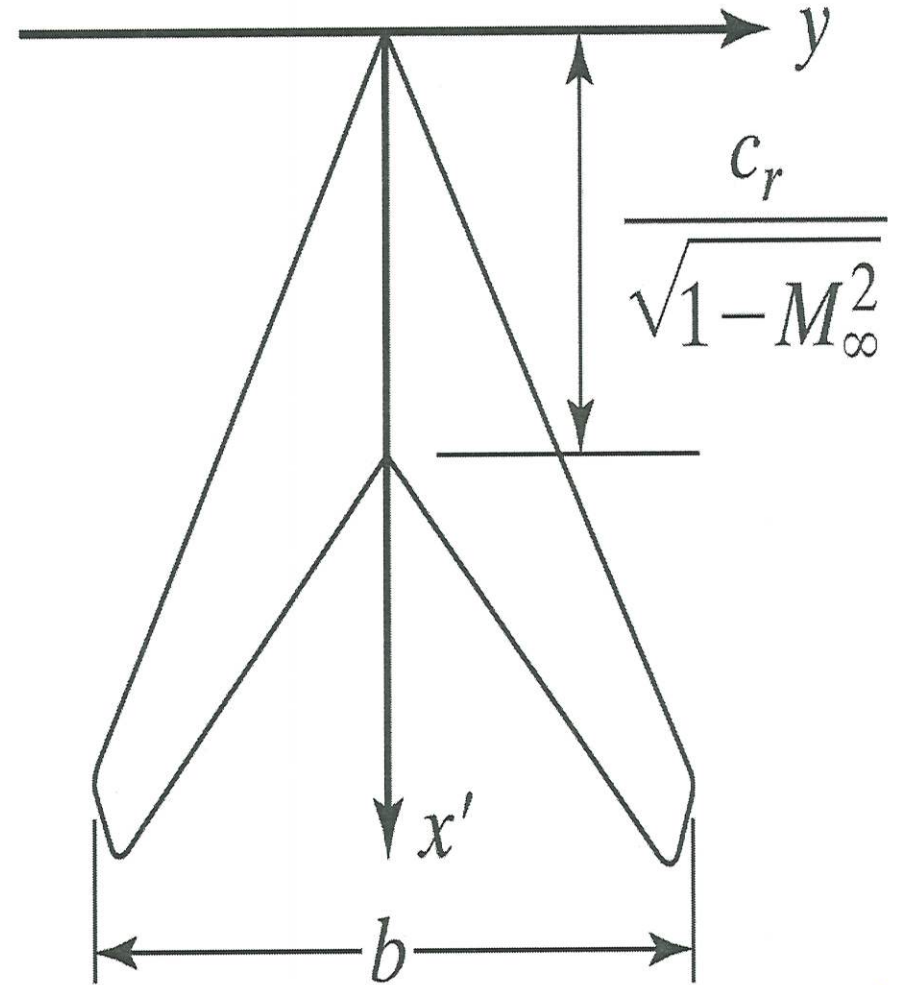
compressible

incompressible



compressible flow

(a)



incompressible flow

(b)

Also:
(integrating C_p)

$$C_l = -\frac{C'_l}{\beta}$$

$$C_m = -\frac{C'_m}{\beta}$$

Compressibility correction to incompressible data (e.g. Abbott and Doenhoff)

d'Alembert's paradox still holds for compressible flow $C_d = 0$
(potential flow cannot predict drag)

Not valid when a shock is present (i.e. transonic flow)

(shock wave drag even for inviscid flow)

Example

the low-speed for NACA 2412 at $\alpha = 4^\circ$ has $C'_\ell = 0.65$ (incompressible)

find C_ℓ at $M_\infty = 0.7$ using Prandtl-Glauert Rule

$$C_l = \frac{C'_l}{\beta} = \frac{C'_l}{\sqrt{1 - M_\infty^2}} = \frac{0.65}{\sqrt{1 - (0.7)^2}} = 0.910$$

For a thin airfoil: $C_\ell = 2\pi\alpha$

Using the Prandtl-Glauert Rule: $C_\ell = \frac{2\pi\alpha}{\beta}$

note: $M_\infty = 0.7 \leadsto \beta = \sqrt{1 - M_\infty^2} = 0.714$

$$C_\ell = C'_\ell / \beta \quad \text{error} \approx 5\%$$

transonic effects

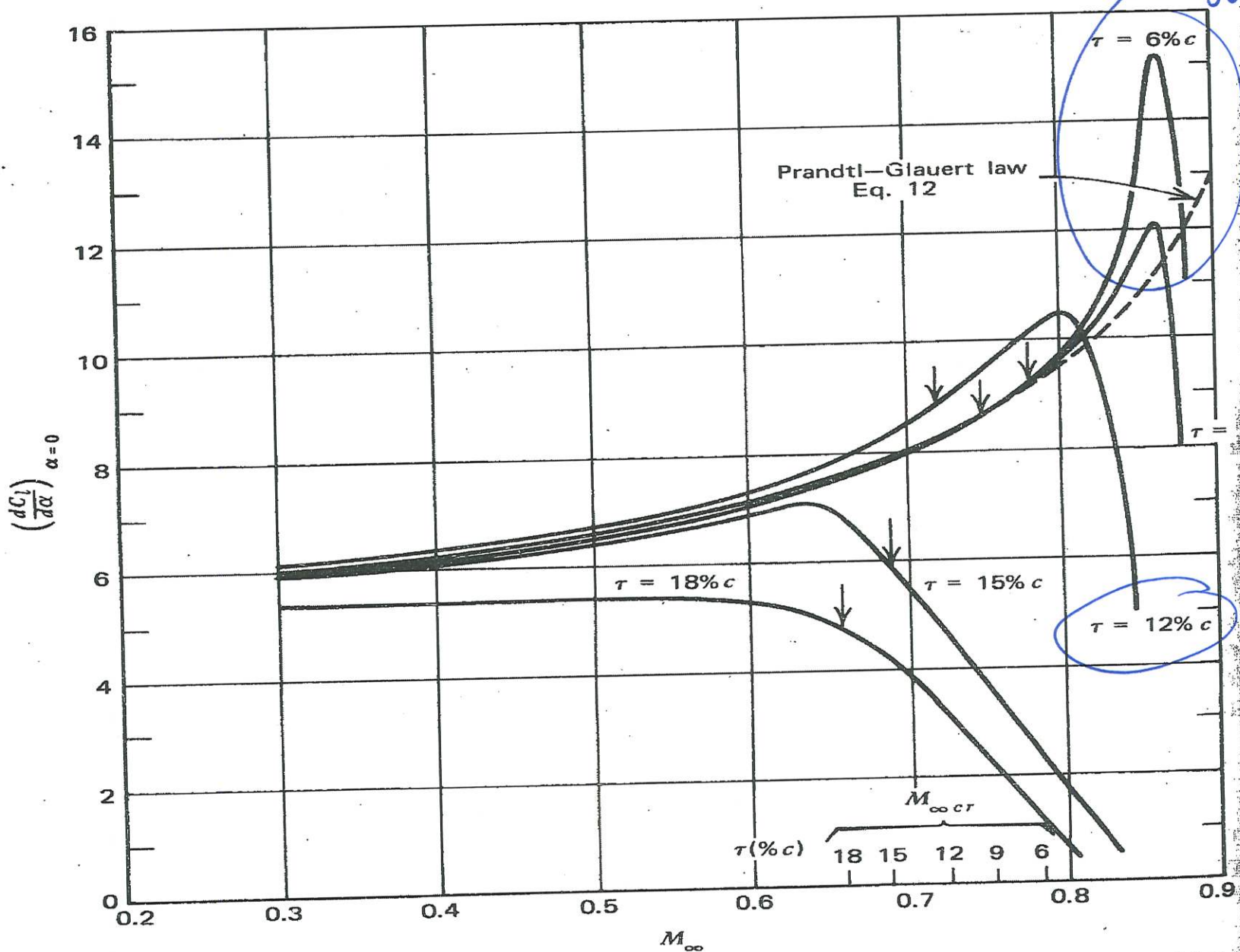
limit
for
P & G
rule

Fig. 4. Measured $dc_l/d\alpha$ versus M_∞ for symmetric airfoils of different maximum thicknesses and comparison with Prandtl-Glauert law for $dc_l/d\alpha = 6.0$ per rad at $M_\infty = 0.3$; $M_{\infty cr}$ each curve is designated by an arrow and by a point on the abscissa. (Mair and Beavan, see 658; Howarth, 1953. Courtesy of Oxford University Press)

$$M_2 = 0.4$$

$$M_2$$

Given data at $M=0.4$ scale data at other Mach numbers

$$\frac{C_{p1}}{C_{p2}} = \frac{C_p' / \sqrt{1 - M_{1\infty}^2}}{C_p' / \sqrt{1 - M_{2\infty}^2}} \Rightarrow C_{p1} = C_{p2} \sqrt{\frac{1 - M_{2\infty}^2}{1 - M_{1\infty}^2}}$$

e.g.

$$M_1 =$$

$$M_2 = 0.4$$

$$0.6$$

$$C_{p1} \approx 1.15 C_{p2}$$

$$0.7$$

$$C_{p1} \approx 1.28 C_{p2}$$

$$0.8$$

$$C_{p1} \approx 1.53 C_{p2}$$

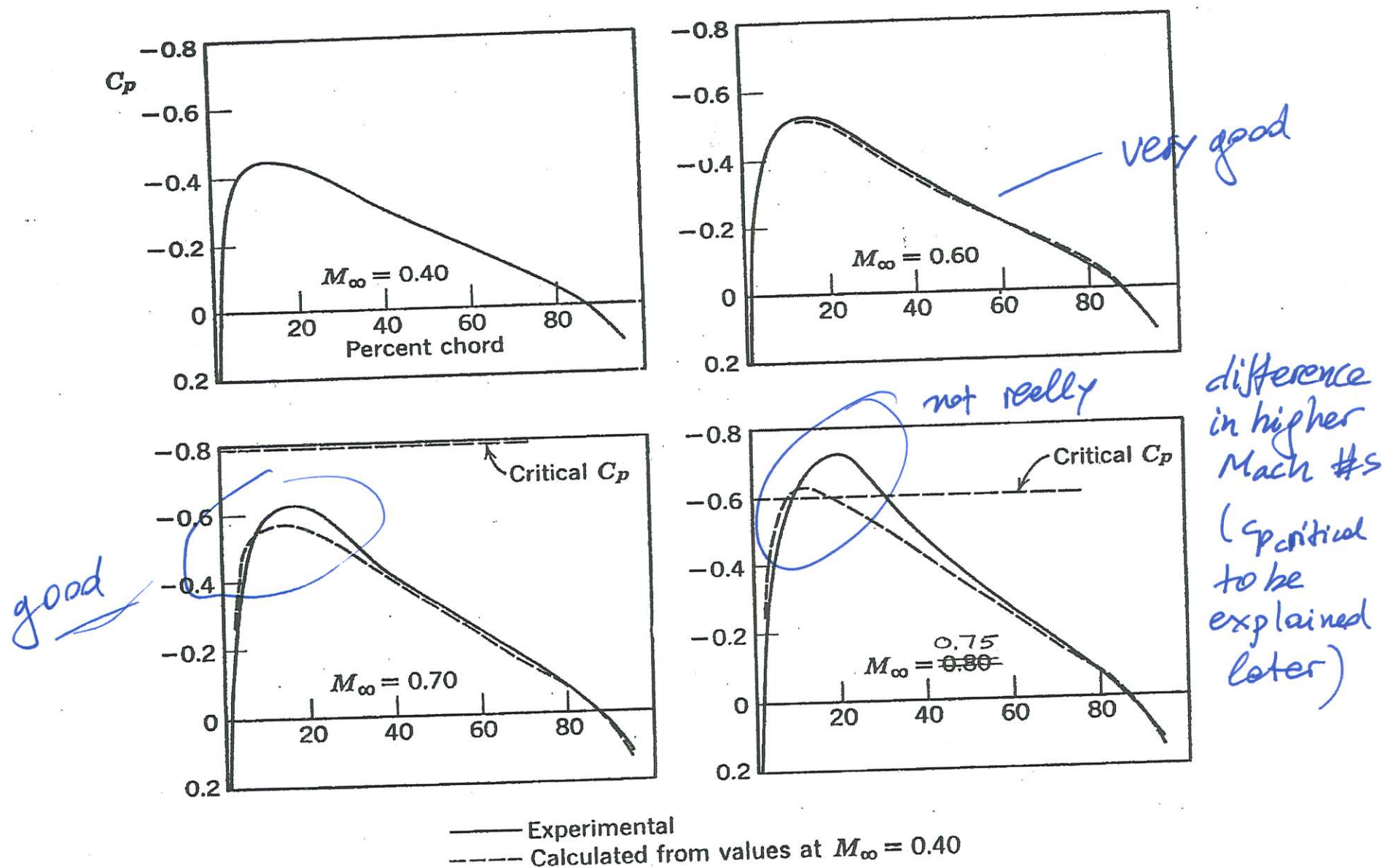


FIG. 10-1 Comparison of Prandtl-Glauert similarity rule with experiment. (Experimental data for NACA 0012 airfoil, taken from *NACA Tech. Note 2174* by J. L. Amick.)

Improved Compressibility Corrections

Some improved compressibility corrections ($M \leq 0.8$)

Karman-Tsien rule:

$$C_p = \frac{C'_p}{\beta + \frac{M_\infty^2}{(1+\beta)} \frac{C'_p}{2}} \quad (9.17)$$

Laitone's rule

$$C_p = \frac{C'_p}{\beta + \frac{M_\infty^2}{\beta} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \frac{C'_p}{2}} \quad (9.18)$$

Both these rules account for some non-linear effects of the flow

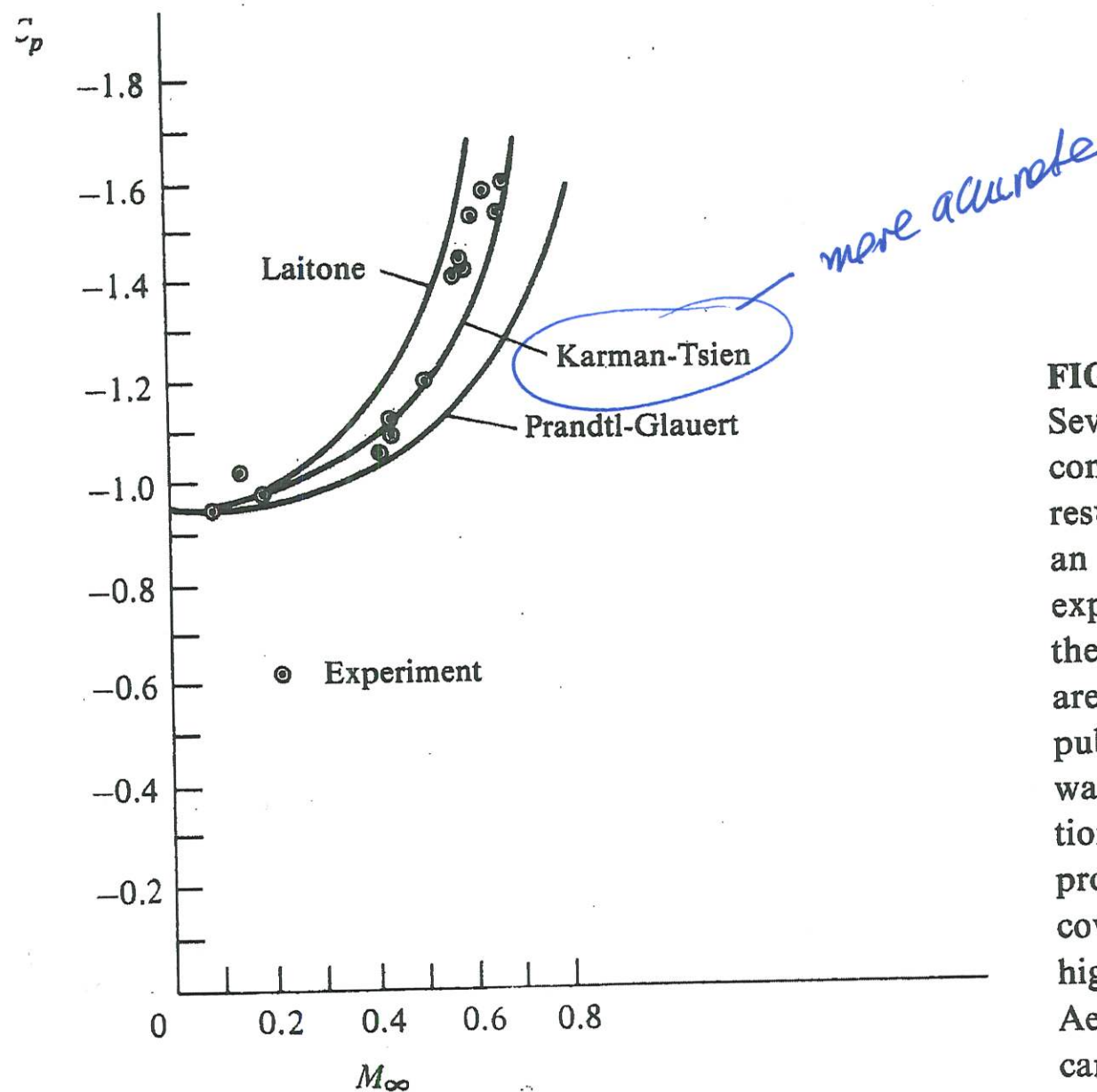
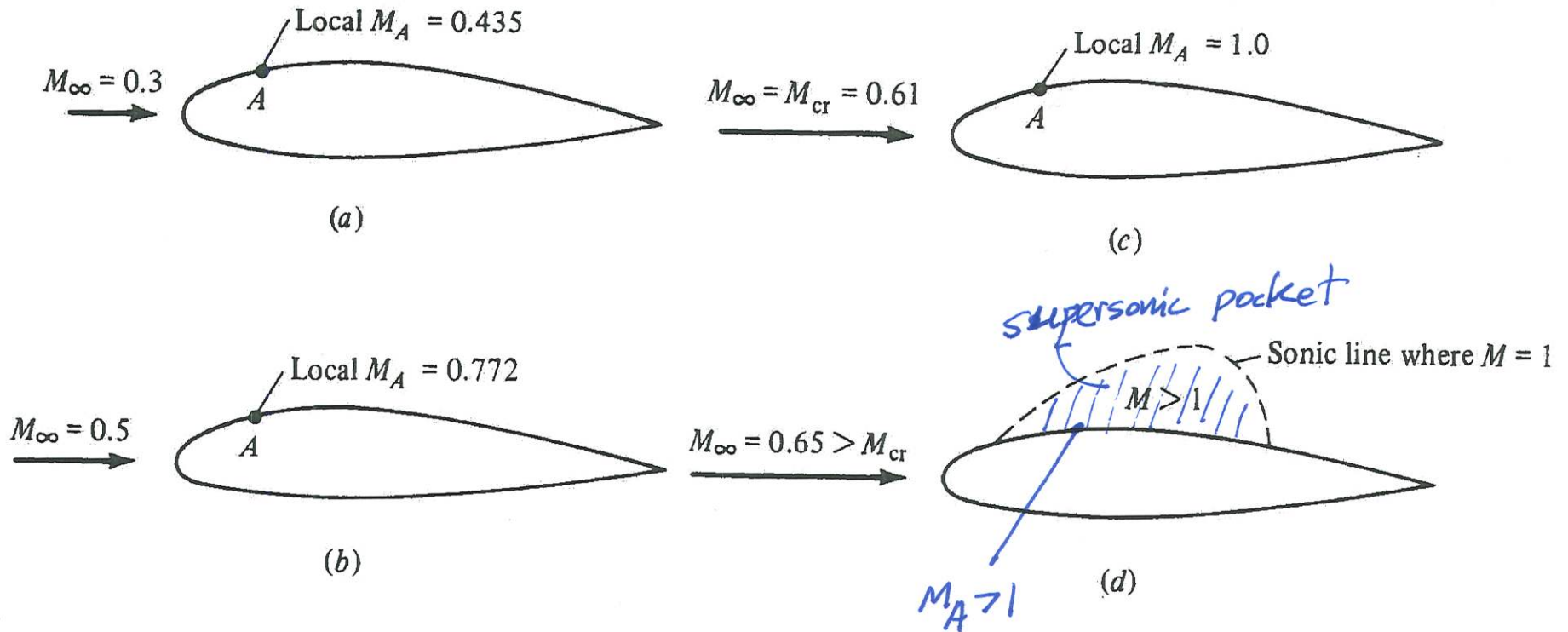


FIGURE 11.4

Several compressibility corrections compared with experimental results for an NACA 4412 airfoil at an angle of attack $\alpha = 1^\circ 53'$. The experimental data are chosen for their historical significance; they are from NACA report no. 646, published in 1938 (Ref. 30). This was the first major NACA publication to address the compressibility problem in a systematic fashion; it covered work performed in the 2-ft high-speed tunnel at the Langley Aeronautical Laboratory and was carried out during 1935-1936.

Critical Mach number

$A = \max M \#$



$M_\infty = 0.61$ is the critical Mach number (~~not~~ Max velocity = $M = 1$)

if $M_\infty > M_{cr}$ increase in drag (transonic)

For isentropic flow p_t constant

$$C_{p,A} = \frac{2}{\gamma M_\infty^2} \left(\frac{p_A}{p_\infty} - 1 \right) \quad (5.47)$$

$\frac{p_{t1}}{p_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\gamma/(\gamma-1)}$
from isentropic flow relations (8.36)

$$\Rightarrow C_{p,A} = \frac{2}{\gamma M_\infty^2} \left[\left(\frac{1 + [(\gamma-1)/2] M_\infty^2}{1 + [(\gamma-1)/2] M_A^2} \right)^{\gamma/(\gamma-1)} - 1 \right] \quad (5.48)$$

for critical Mach number: $M_A = 1$

$$\rightarrow C_{p_{cr}} = \frac{2}{\gamma M_{cr}^2} \left[\left(\frac{1 + [(\gamma-1)/2] M_{cr}^2}{1 + (\gamma-1)/2} \right)^{\gamma/(\gamma-1)} - 1 \right] \quad (5.49)$$

this equation can be used for all airfoils

M_{cr}	$C_{p_{cr}}$	as $M \uparrow$ $C_{p_{cr}} \downarrow$	
0.4	-3.66		
0.6	-1.29		
0.7	-0.78		
0.75	-0.59		
0.8	-0.43		

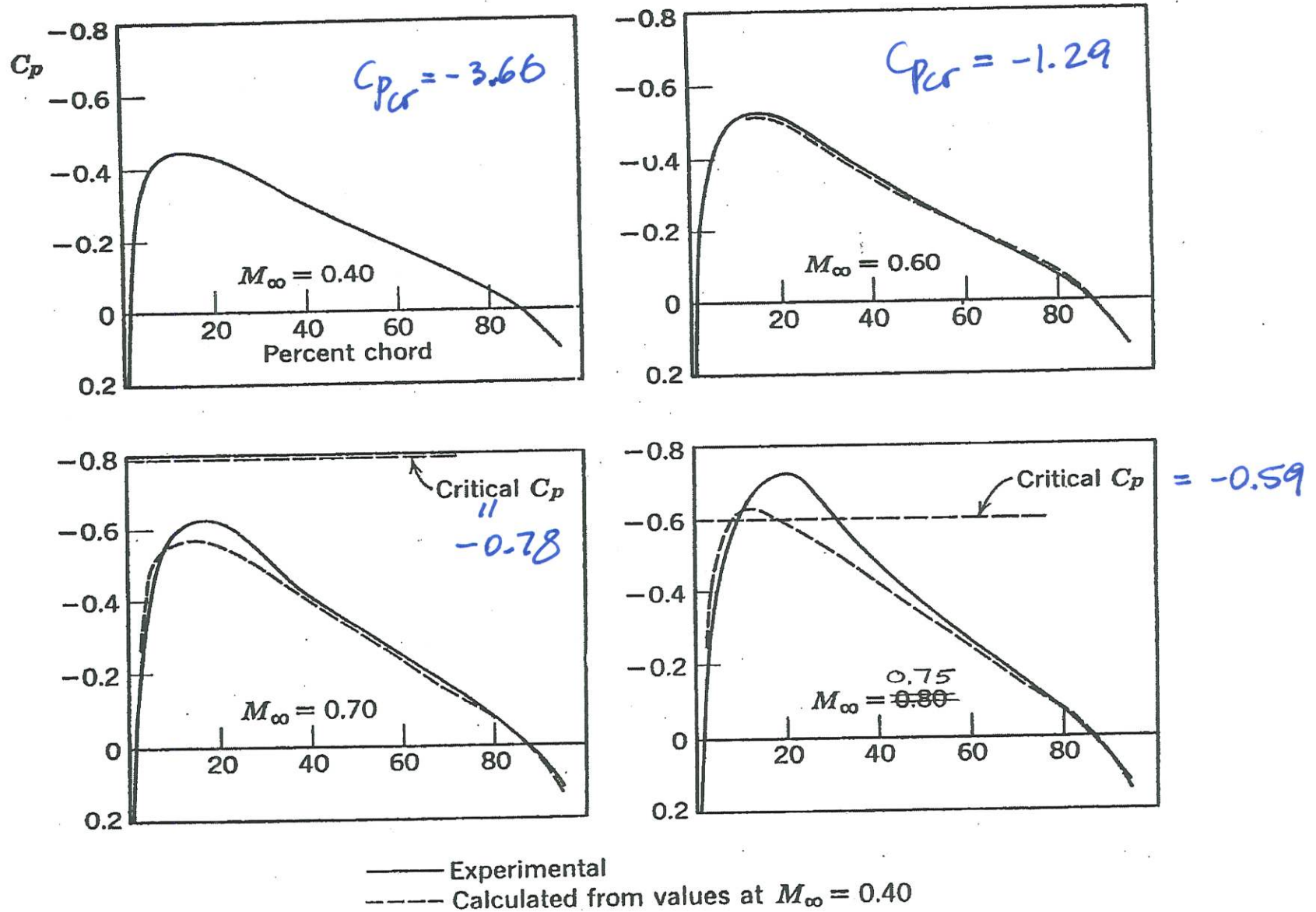
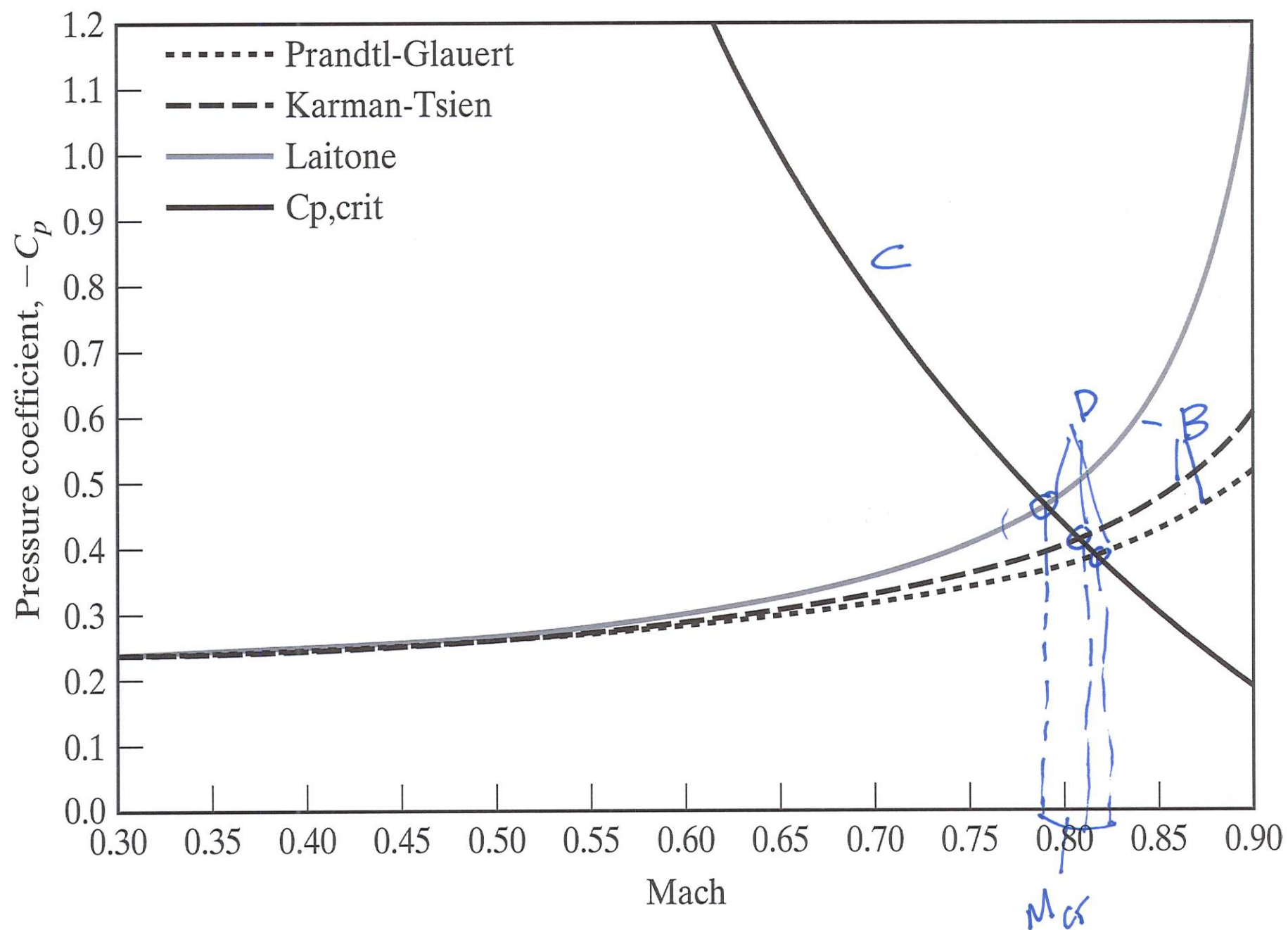


FIG. 10-1 Comparison of Prandtl-Glauert similarity rule with experiment. (Experimental data for NACA 0012 airfoil, taken from *NACA Tech. Note 2174* by J. L. Amick.)



M_{cr} is lower for thicker airfoils; but needs to be high in airplanes

$\Rightarrow$ thin airfoils e.g. Gates-Lear – 9% thick

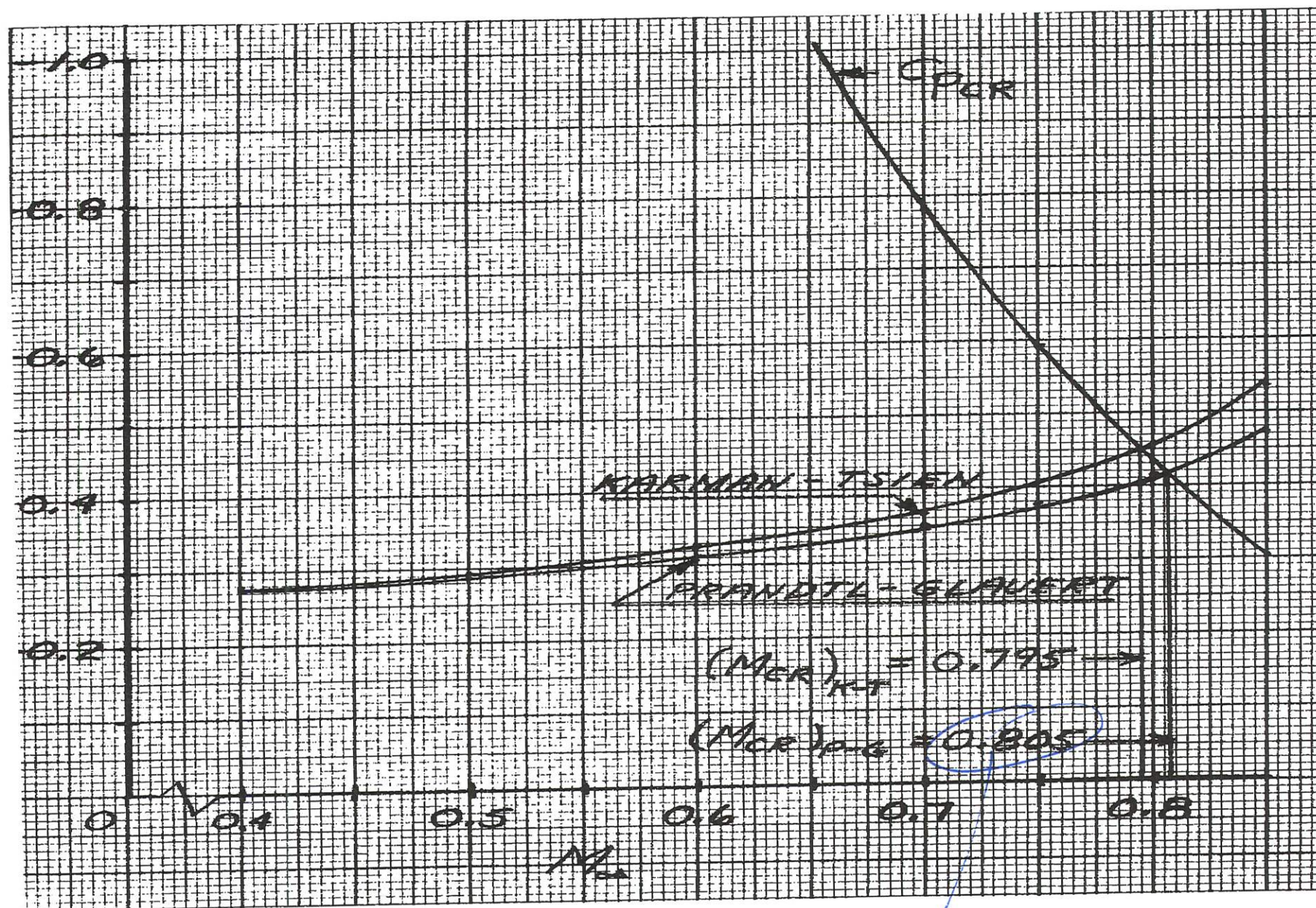
Piper Aztec 14% thick (low speed)

Example

At $\alpha = 0^\circ$ the incompressible min pressure coefficient for a NACA
0009 airfoil is -0.25 . ^{c_p} Calculate the M_{cr} for this airfoil using

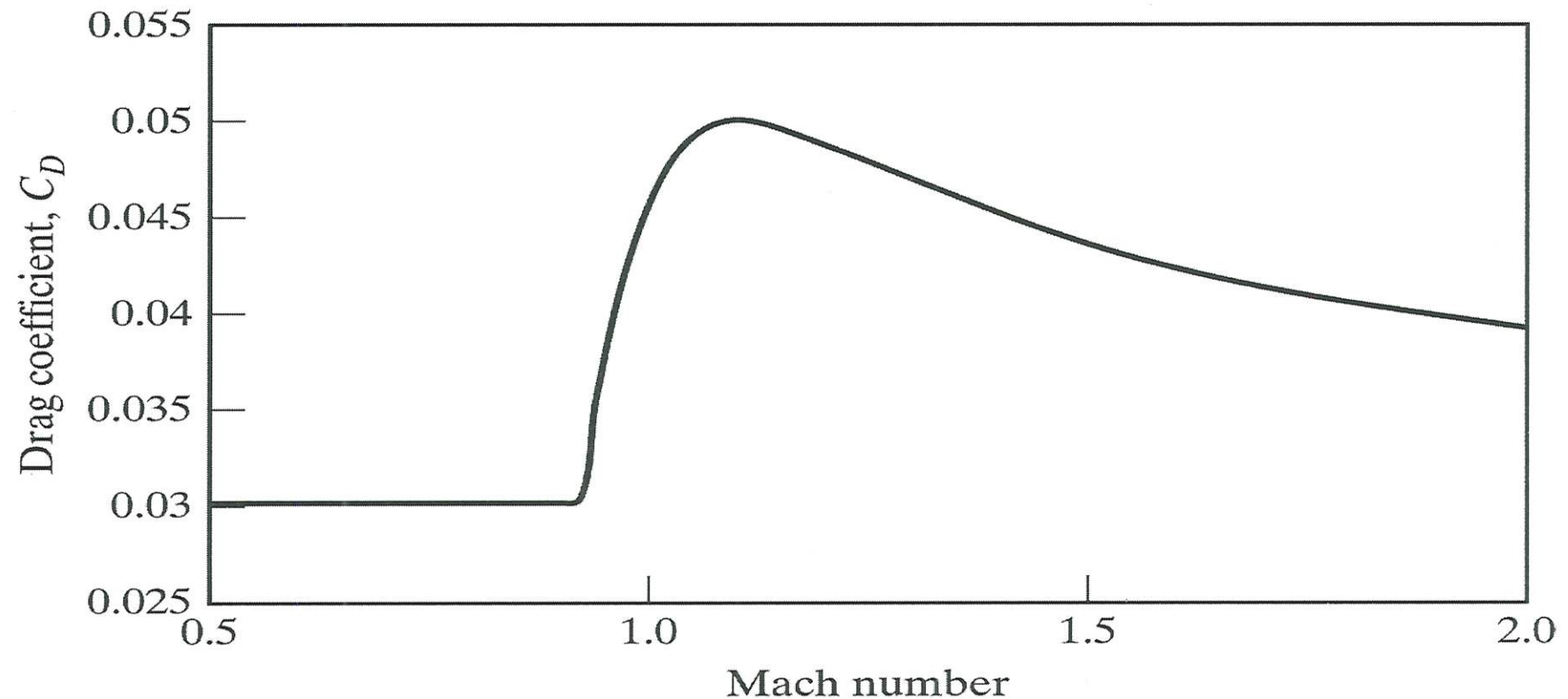
- a) Prandtl-Glauert Rule
- b) The (more accurate) Karman-Tsien rule

M_∞	$C_{p_{cr}}$ eq. (9.21)	$C_{p_{Pr andtl-Glauret}}$	$C_{p_{Karman-Tsien}}$
0.3	- 6.9	- 0.262	- 0.264
0.4	- 3.66	- 0.273	- 0.276
0.5	- 2.13	- 0.289	- 0.294
0.6	- 1.29	- 0.313	- 0.323
0.7	- 0.78	- 0.35	- 0.369
0.75	- 0.59	- 0.378	- 0.404
0.80	- 0.43	- 0.417	- 0.455
0.85	- 0.302	- 0.475	- 0.535



a little higher

Drag Divergence Mach Number

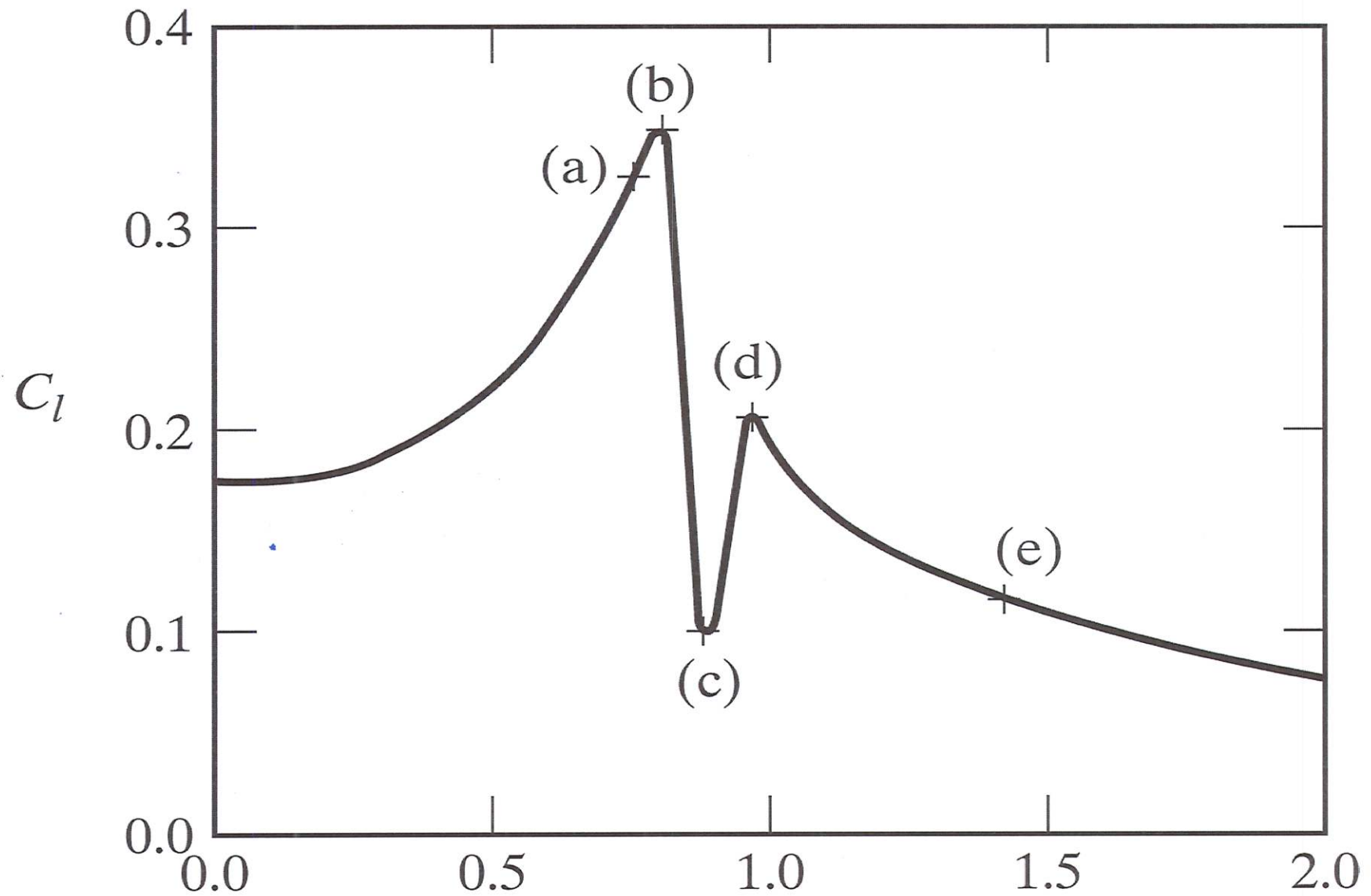


M_{dd} either 2-4% higher than M_{cr} , or $dC_D / dM_\infty = 0.03, 0.05, 0.10$

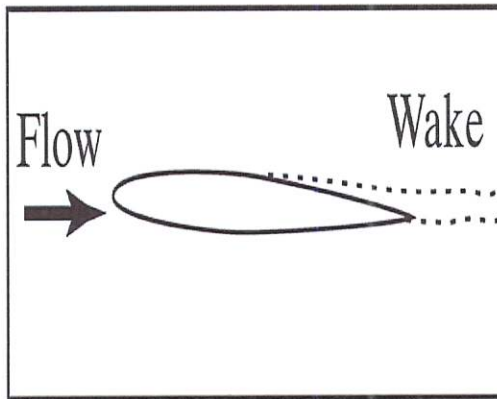
$$M_{dd} = M_{cr} \left[1.02 + 0.08(1 - \cos \Lambda) \right] \quad (9.22)$$

sweep angle

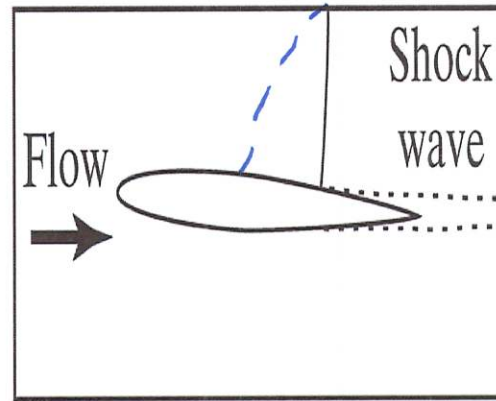
empirical approximation based on $dC_D / dM_\infty = 0.05$



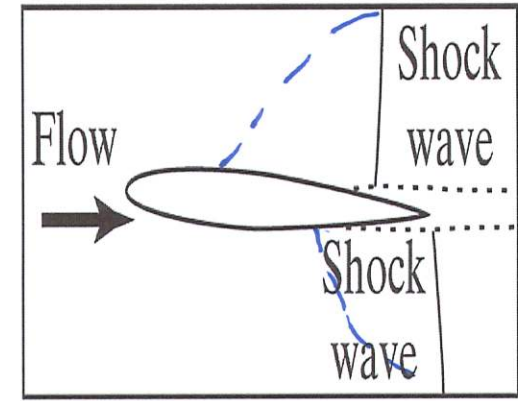
lift coefficient as a function of M



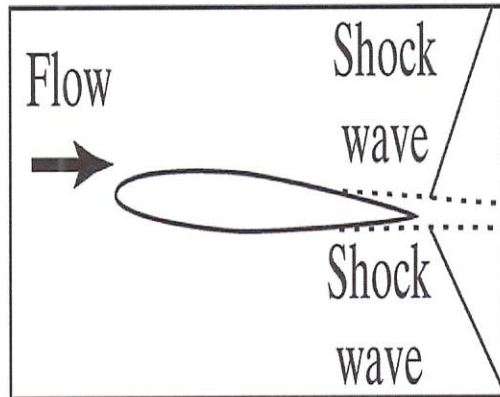
(a)



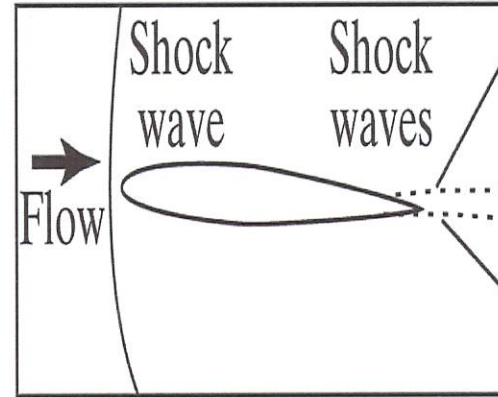
(b)



(c)



(d)

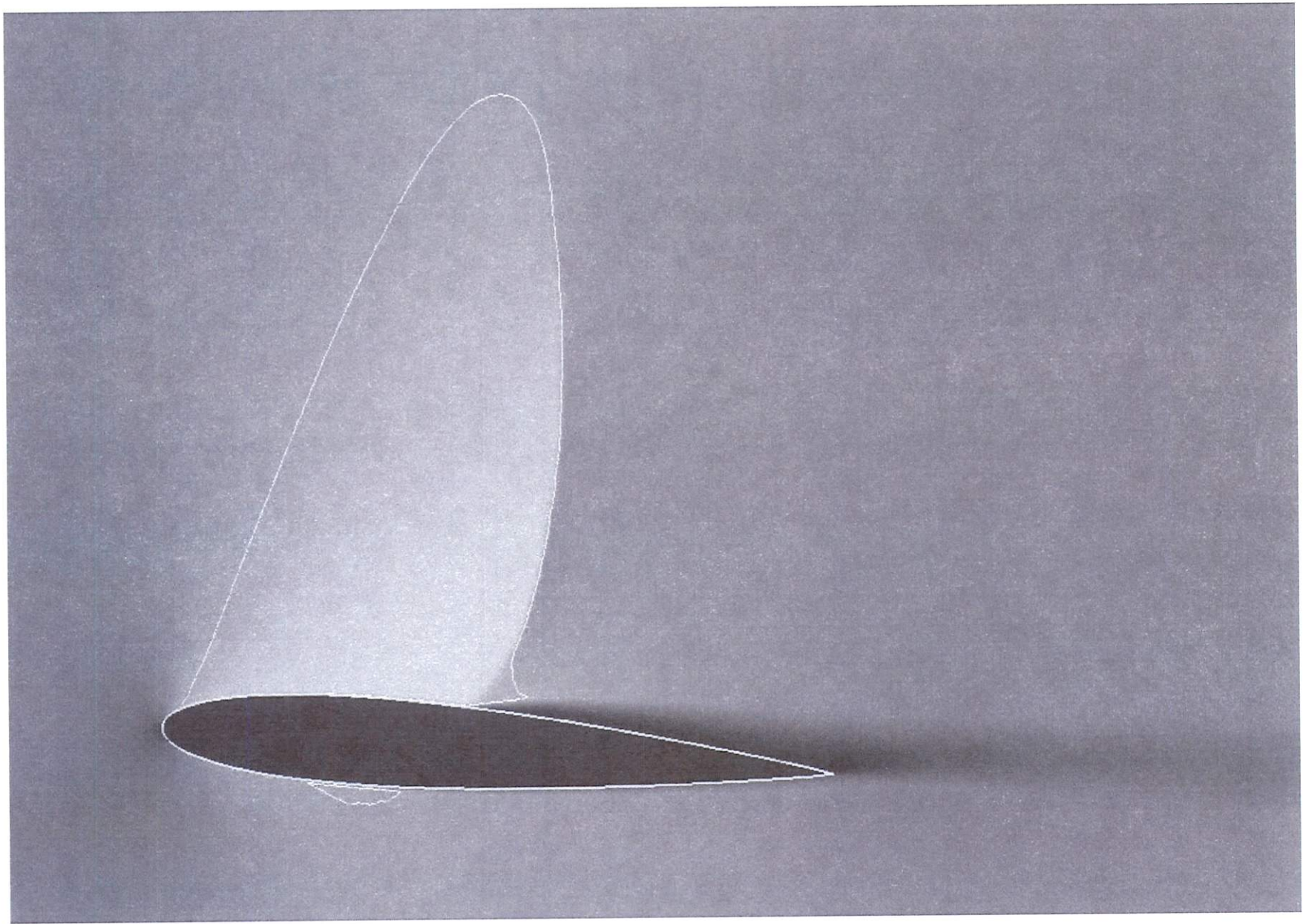


(e)

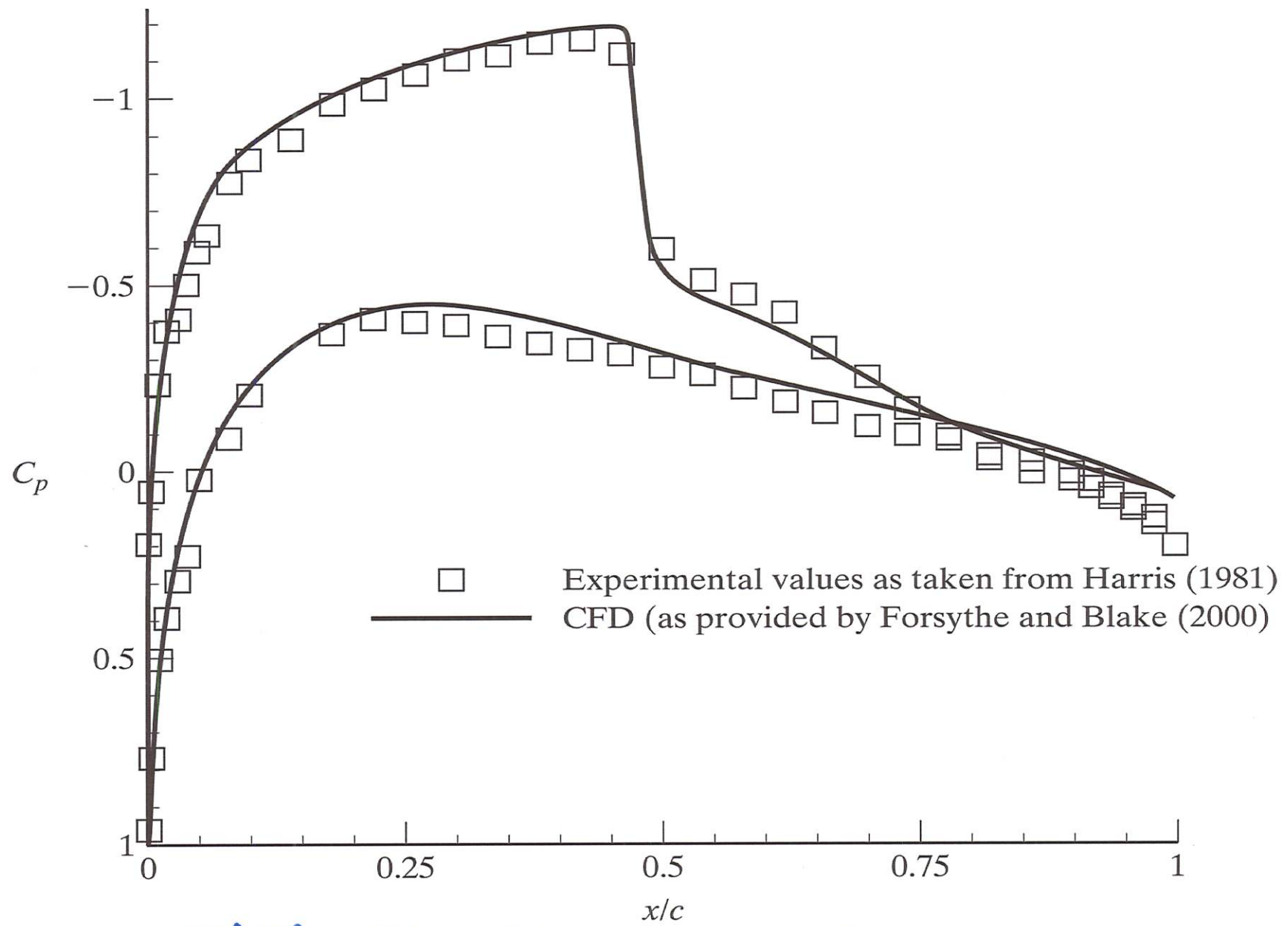
increasing M :

32

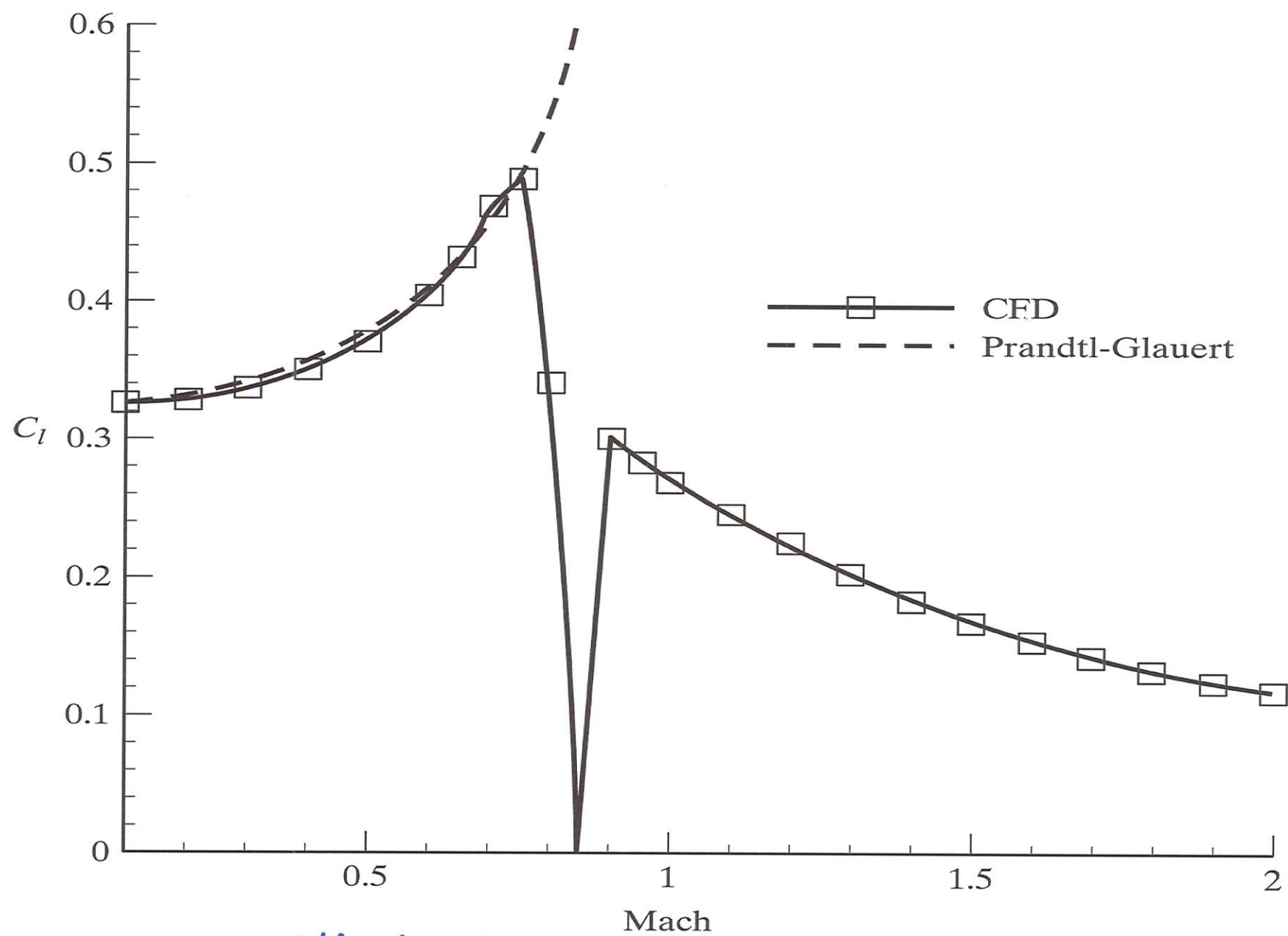
- a) $M_\infty = 0.75$ subcritical
- b) $M_\infty = 0.81$ supercritical on the top, shock wave maximum lift
- c) $M_\infty = 0.89$ shock in upper surface becomes stronger; shock in lower surface
- d) $M_\infty = 0.98$ both shocks reach trailing edge
- e) $M_\infty > 1$ detached shock



NACA 0012, $M = 0.8$, $\alpha = 3^\circ$



NACA 0012, $M=0.8$, $\alpha=3^\circ$

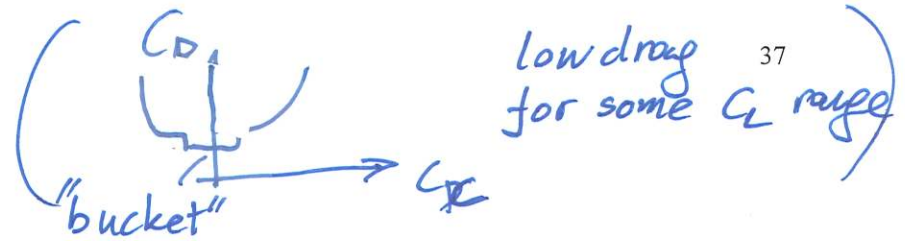


NACA 0012



B-1B transonic flight (condensation & evaporation of water)

Supercritical airfoils



- i) increase M_{cr} $\rightarrow$ thin airfoils (1945-1965) NACA 64 series
(originally developed to encourage laminar flow) need thick airfoil for: strength - fuel storage
- ii) increase the Mach number increment between M_{cr} and M_{dd}

1945-1965- NACA 64 series

1965 $\rightarrow$ now $\Rightarrow$ supercritical airfoils

used in the 80's

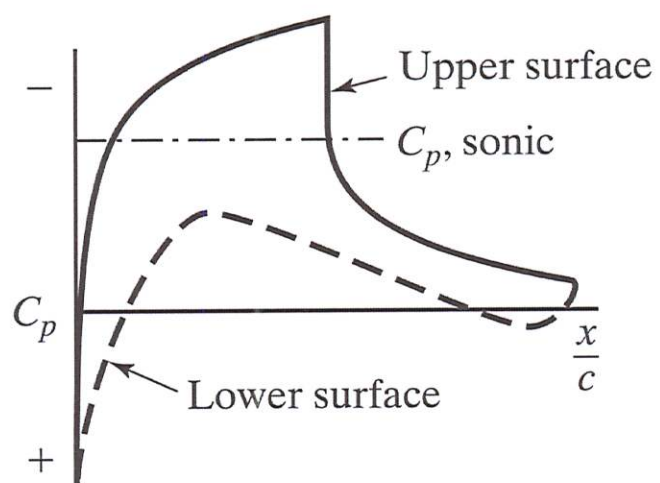
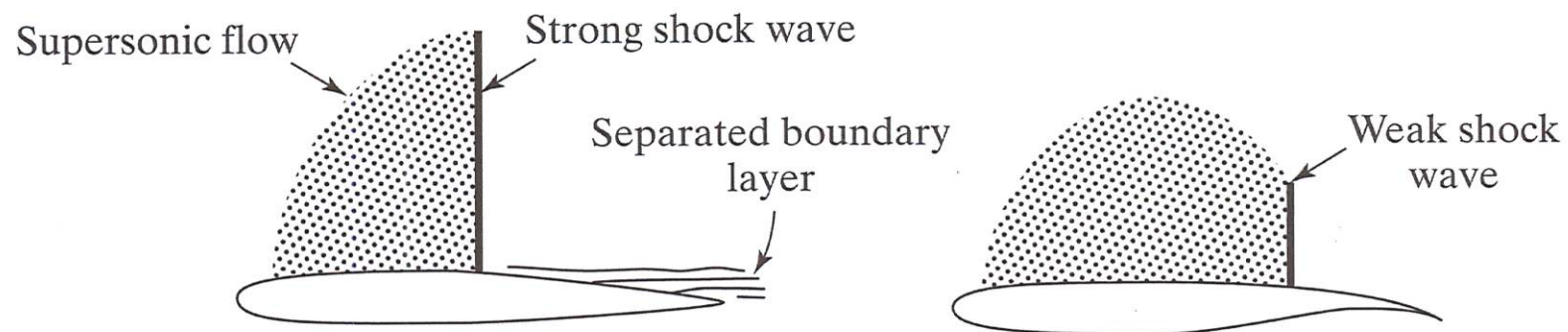
Regular airfoils have a strong shock (separation, high drag)

Supercritical airfoils have a flatter top; shock further back, weaker, mild or no separation.

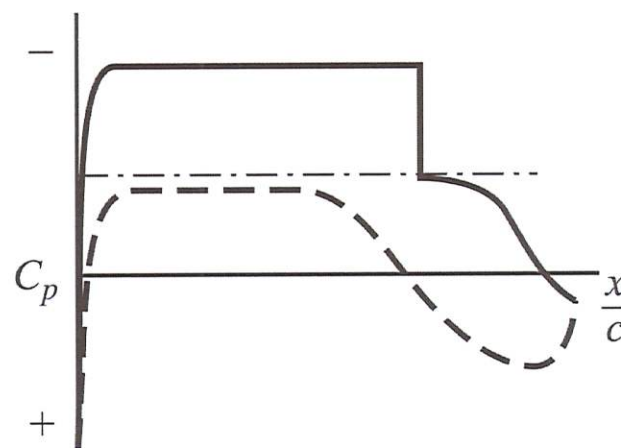
Off-design characteristics suffer lift and drag penalties *(later release)*

Question: 757, 767, 777 use supercritical airfoils; 717 did not. why?

Answer: cost (used earlier wing from an MD airplane)



(a)



(b)

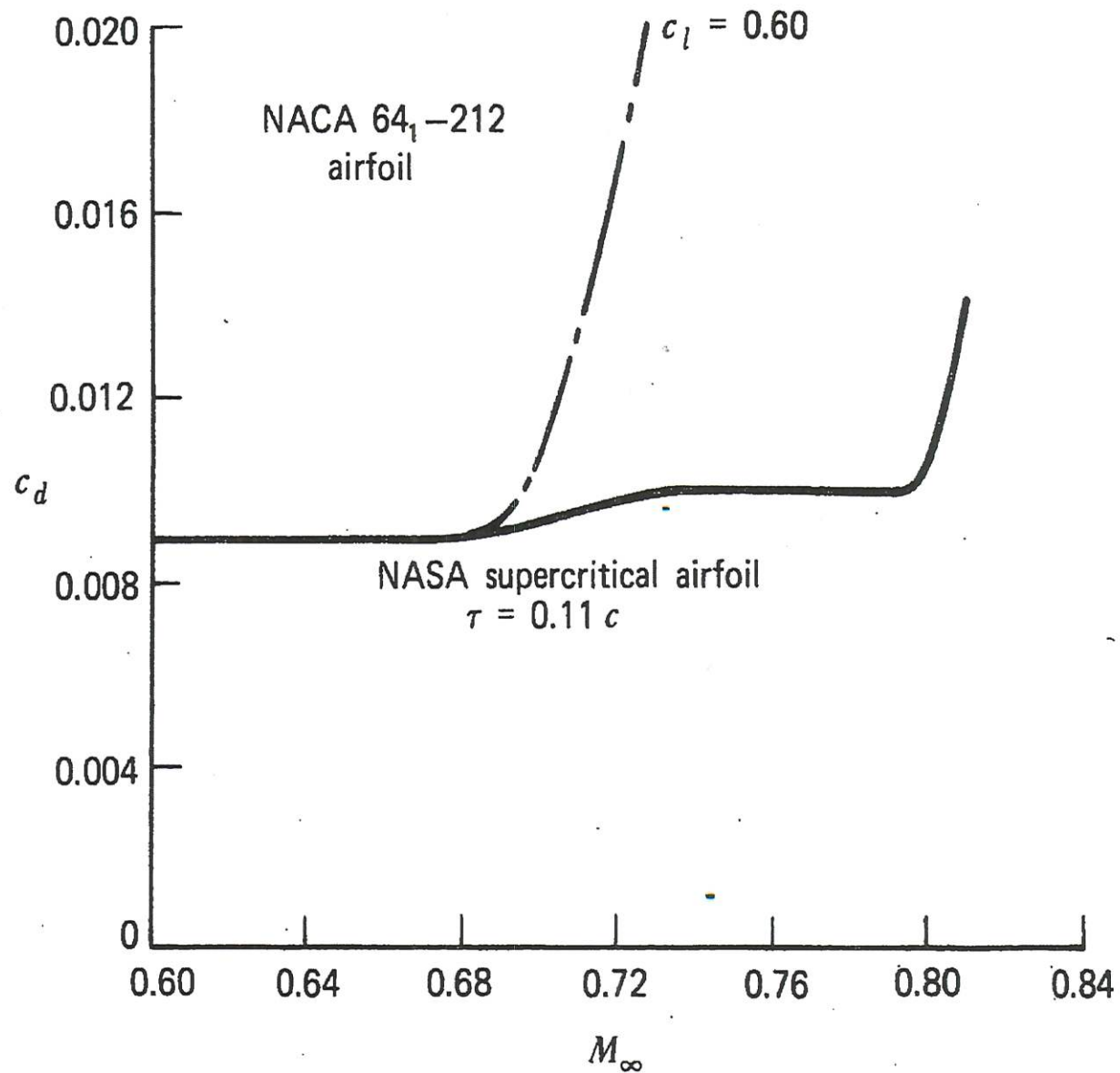


Fig. 8. Characteristics of NASA supercritical airfoil compared with a conventional section. (Whitcomb, 1956. *Courtesy of NASA*)

Swept Wings

Independence Principle: flow can be split in normal + tangential

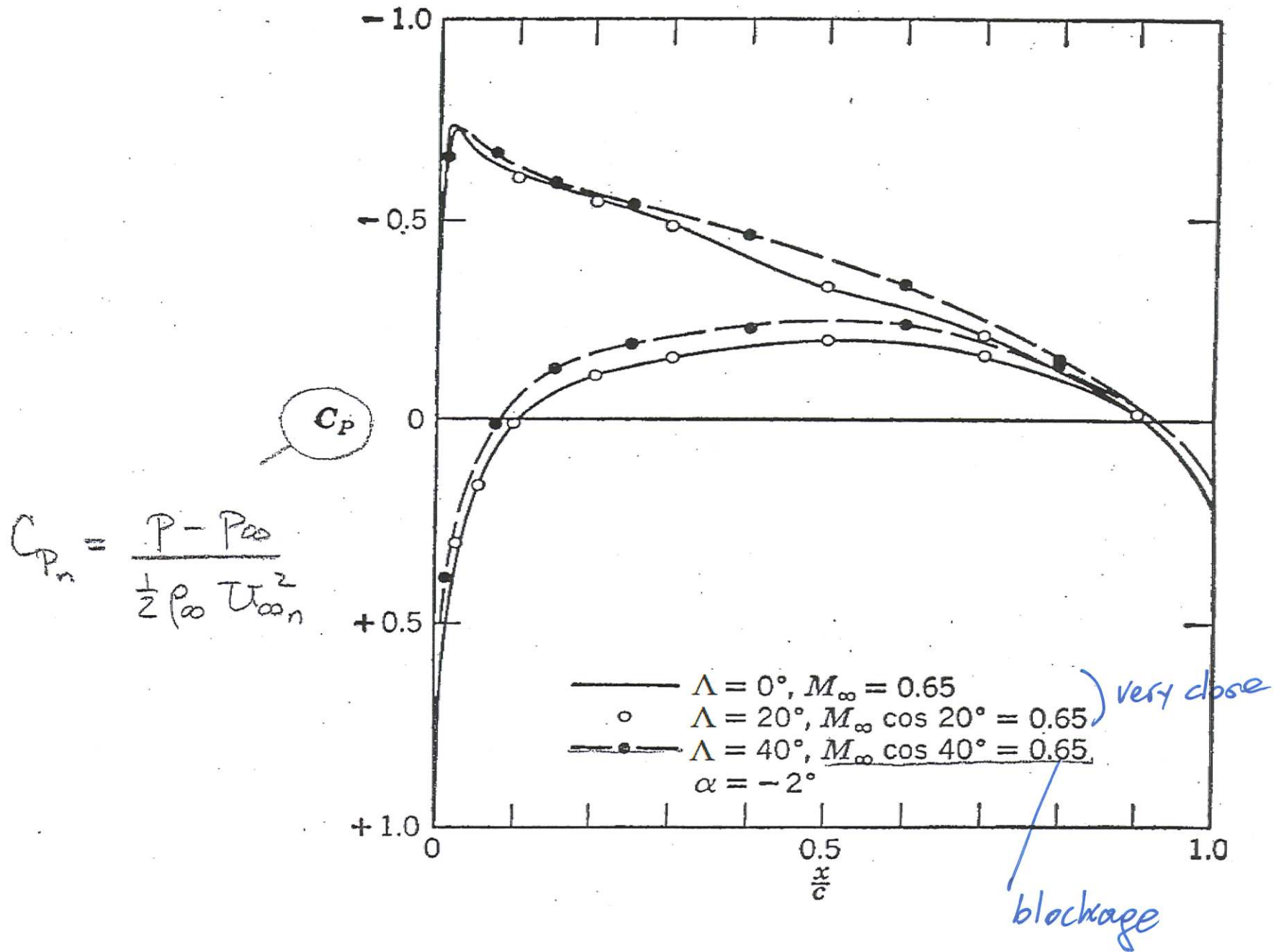
Assume a sweep angle Λ

The normal Mach number is: $M_{n, \infty} = M_{\infty} \cos \Lambda$

The point when the flow becomes supersonic or goes above drag divergence is delayed; any shock is weaker or delayed

The critical normal Mach number: $M_{n, cr} = M_{\infty, cr} \cos \Lambda$

The pressure coefficient is: $C_{p n} = \frac{p - p_{\infty}}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 \cos^2 \Lambda} = \frac{\frac{p}{p_{\infty}} - 1}{\frac{1}{2} \gamma M_{n, \infty}^2}$



Forward Swept Wings

Structural issues; use aeroelastic tailoring and composites

Reduced shock wave drag, increased maneuverability
(important for fighter airplanes)

Higher AR for the same bending moment

Stall starts at root (early warning, maintain aileron effectiveness)

Static instability



NASA Dryden Flight Research Center Photo Collection
<http://www.dfrc.nasa.gov/Gallery/Photo/index.html>
NASA Photo: EC87-0182-14 Date: July 24, 1987 Photo By: NASA

X-29 in Banked Flight

Transonic Area Rule: Area varies smoothly

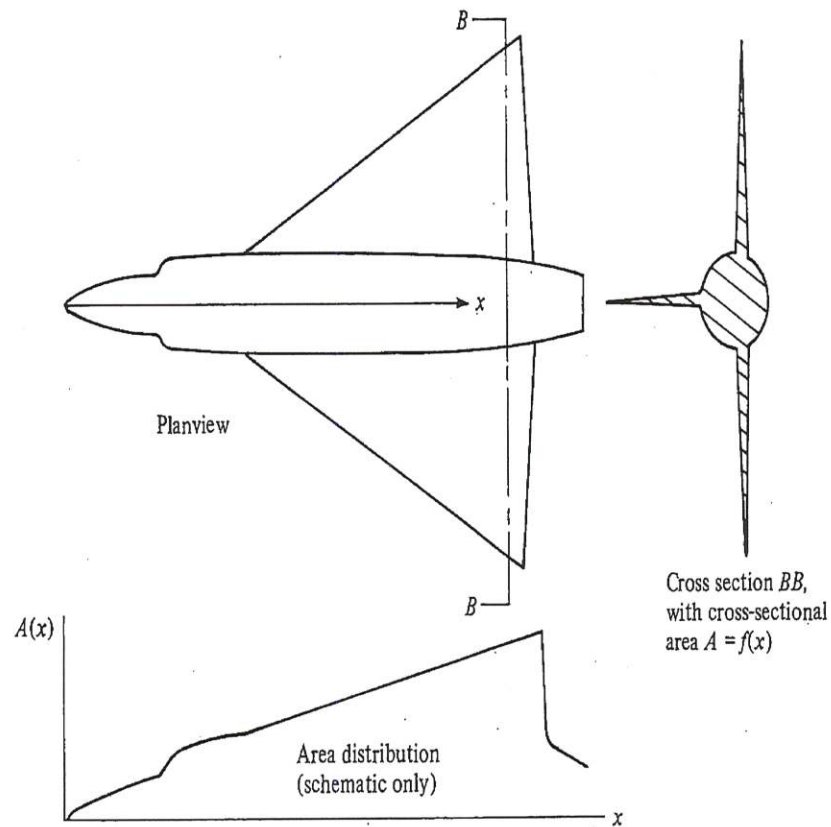


FIGURE 11.10
A schematic of a non-area-ruled aircraft.

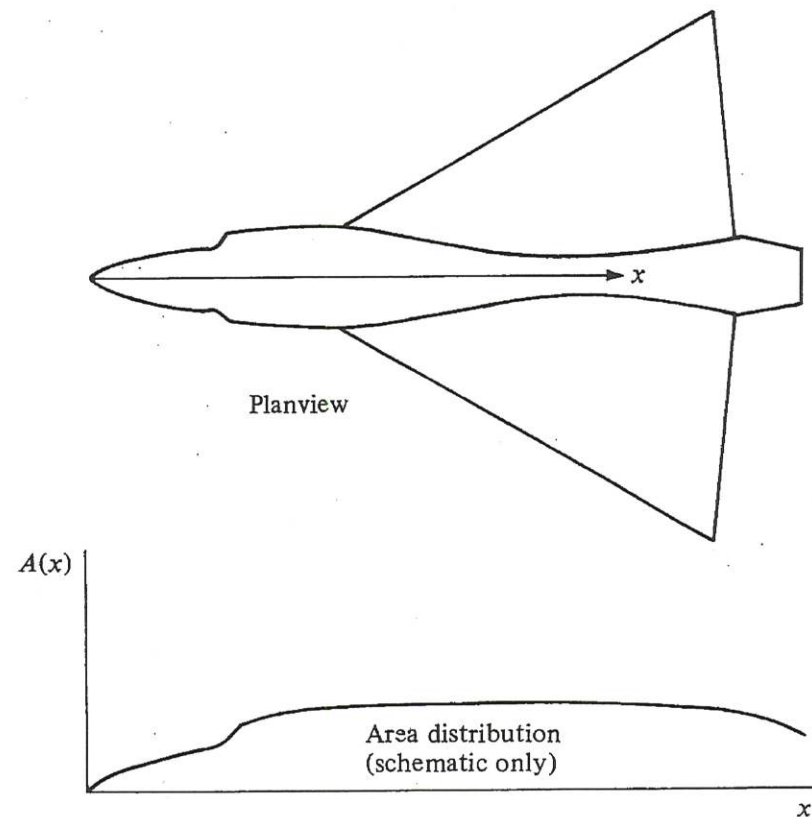
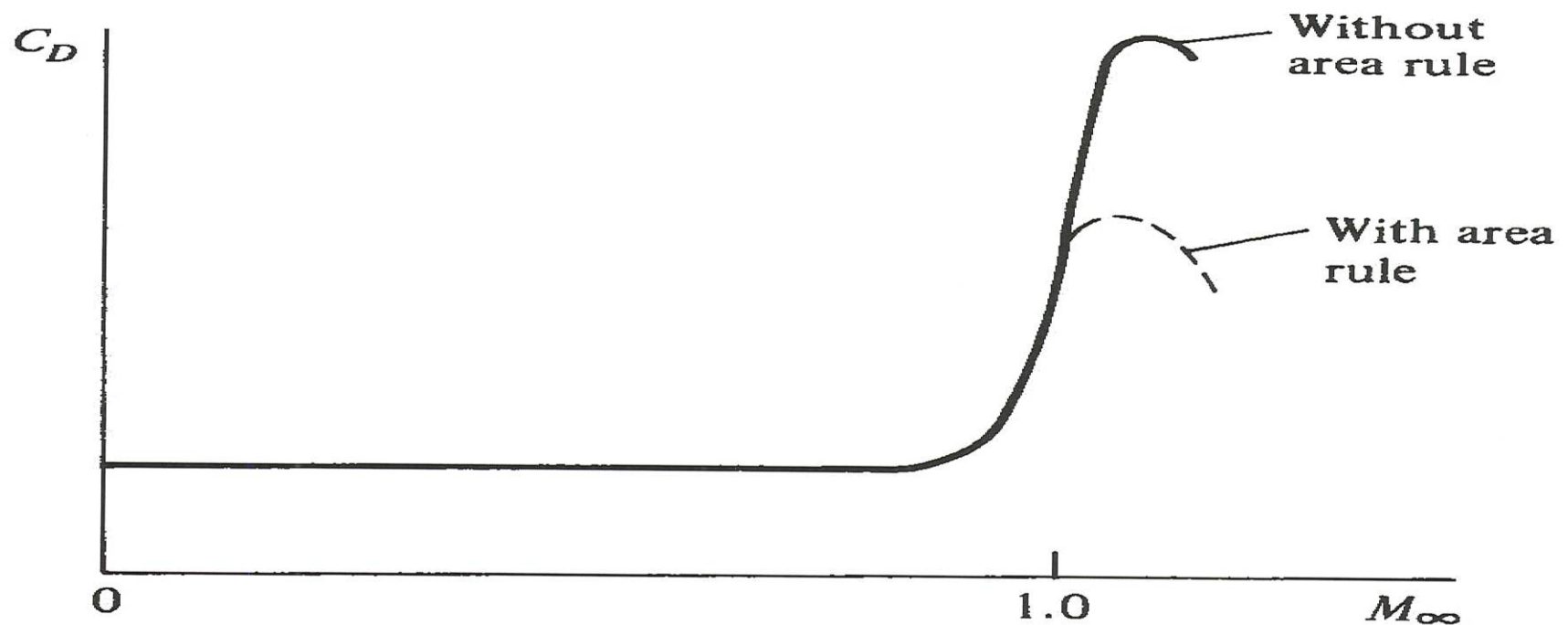


FIGURE 11.11
A schematic of an area-ruled aircraft.

ballistics: speed of supersonic bullets is better for smooth variation of cross-sectional area

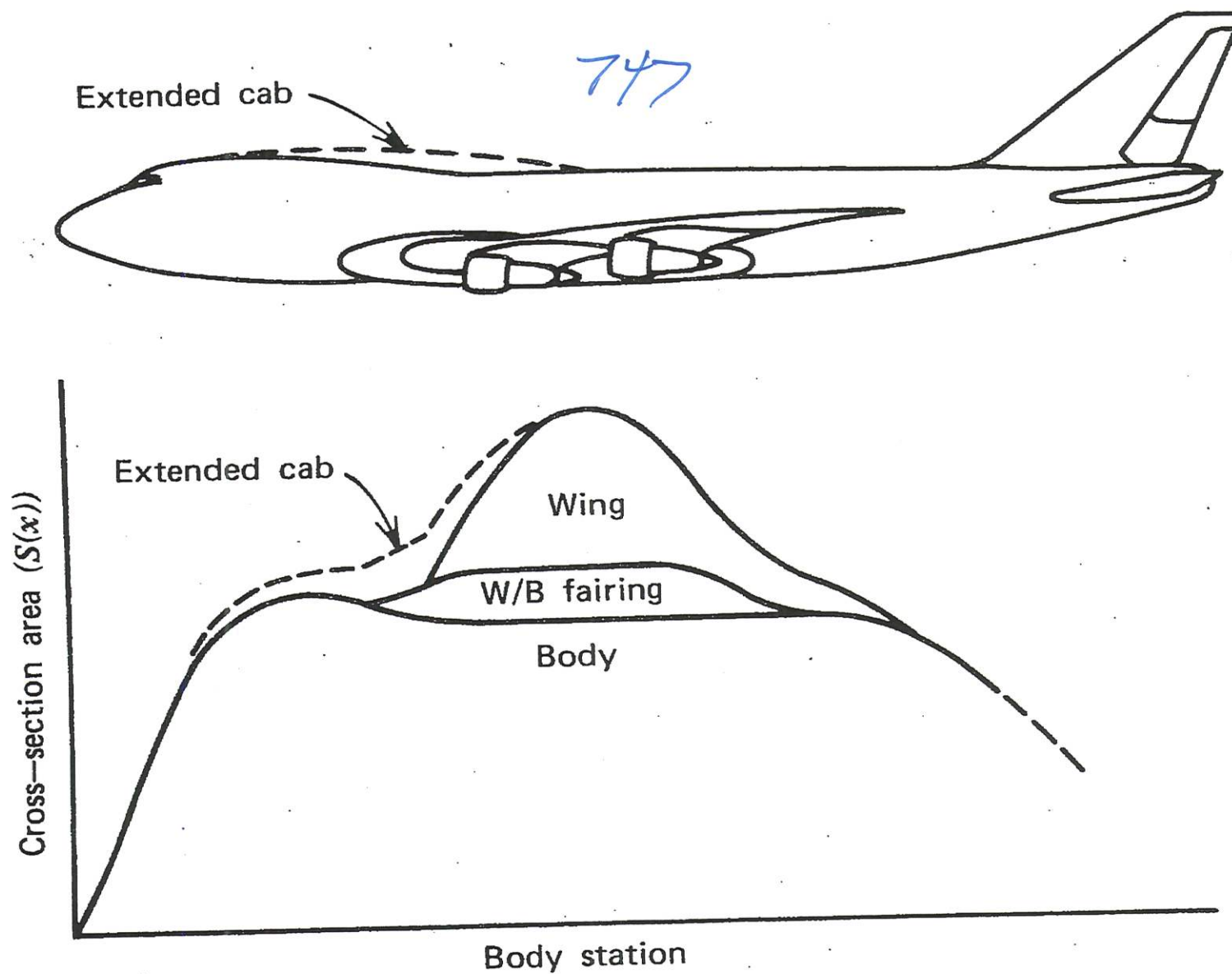


Whitcomb developed the rule in the mid 1950's



Photo Copyright © Air2Air

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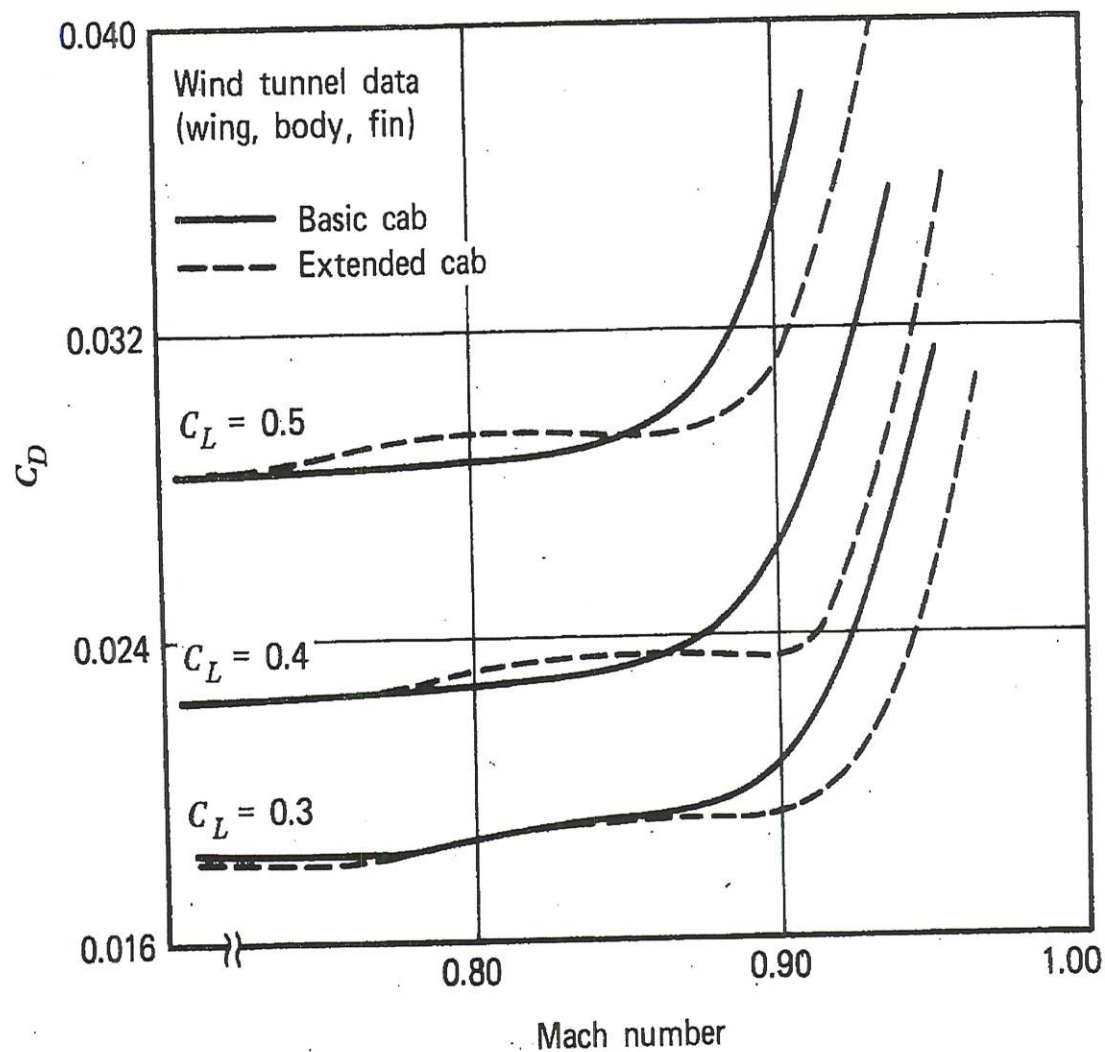


Fig. 9. Effect on $S(x)$ and on measured drag of Boeing 747 due to fuselage modification. (Goodmanson and Gratzner, 1973. *Courtesy of the Boeing Company*)

Transonic aircraft

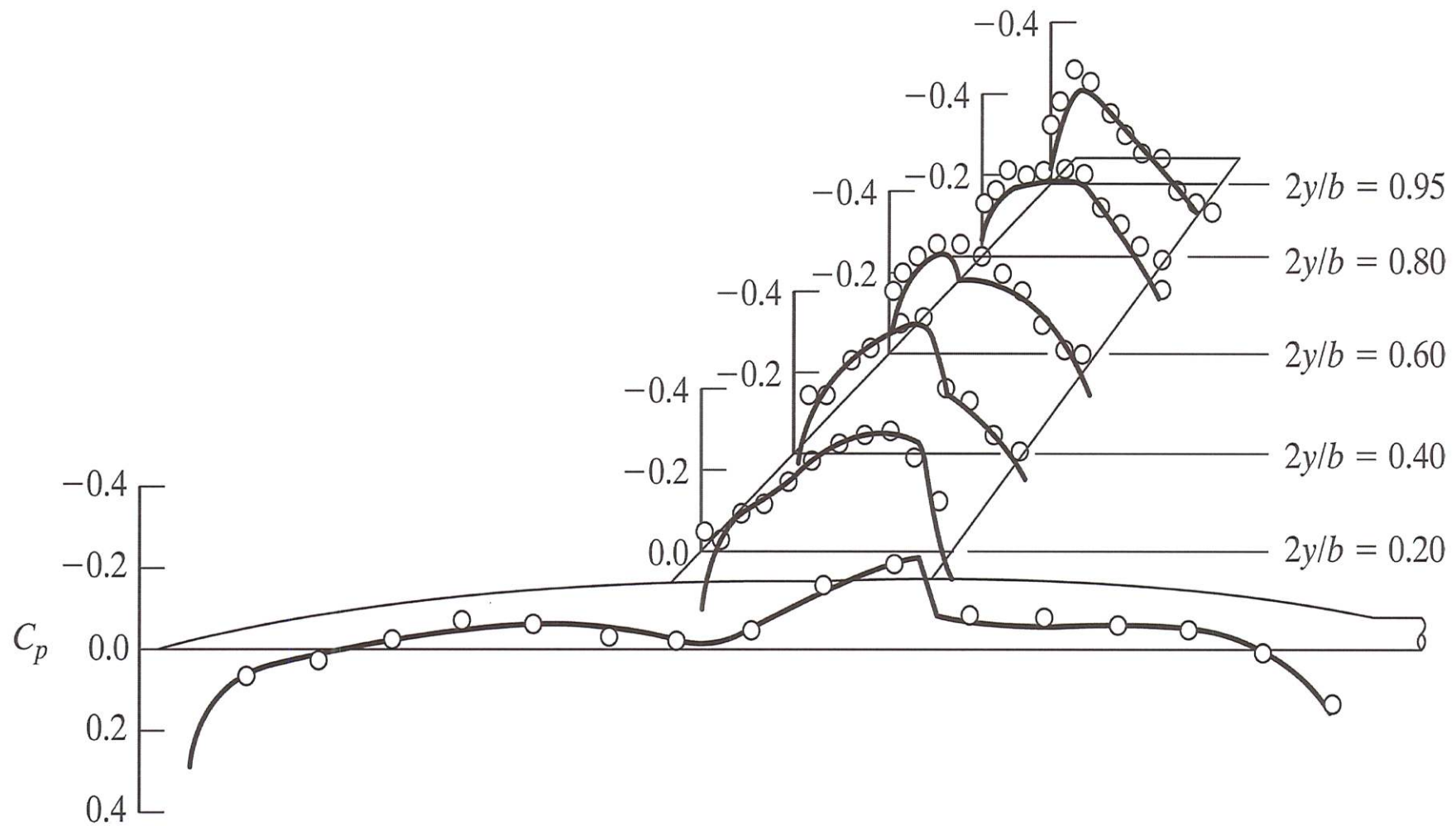
Full potential equation (Eq. 9.4b) has been used. *extensively*
(flow over wings)
 Faster is the transonic small disturbance (TSD) equation

$$\beta^2 \phi_{xx} + \phi_{yy} = \frac{2}{\gamma} \left(1 + \frac{\gamma - 1}{2} M_\infty^2 \right) M_\infty^2 \phi_x \phi_{xx} \quad (9.23)$$

Other versions of this equation are available (see text)

Euler &

Navier Stokes solvers are increasingly used.



C_p for swept-wing/fuselage configuration, $M_\infty = 0.93$; $\alpha = 0^\circ$,
 $\Lambda_{c/4} = 45^\circ$, $AR = 4$, $\lambda = 0.6$, NACA 65A006 streamwise section.

small disturbance eq. Bailey Bellhaus 1975